

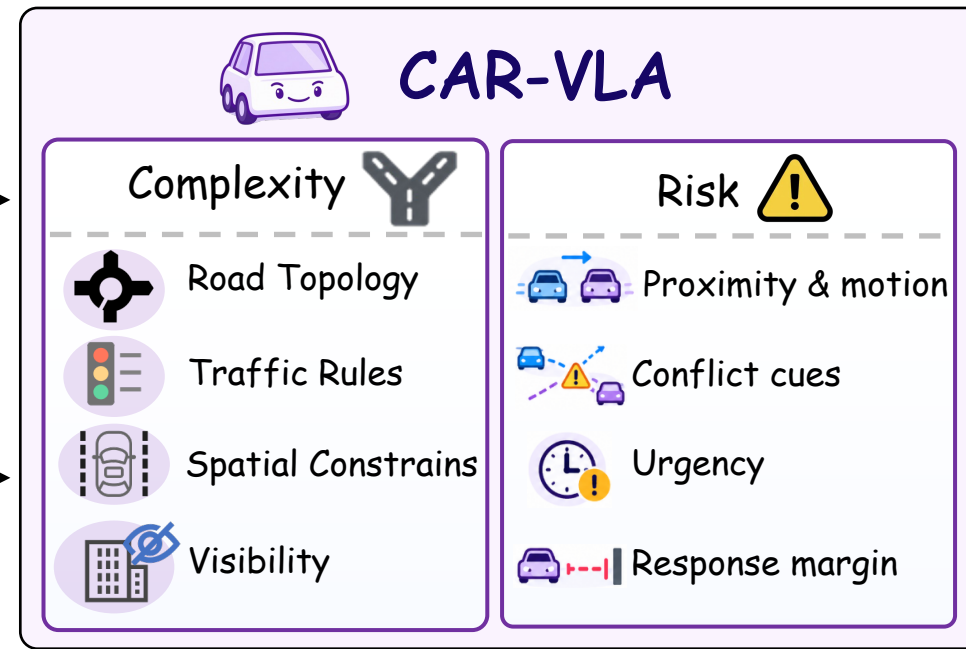
(a) CAR-VLA Overview

Multi-view images



Vision Encoder

Text Tokenizer



Simple Low risk

Complex Low risk

Simple High risk

Complex High risk



Fast Intuition



Slow Thinking

- What do I see?
- Which object is crucial?
- What is the Ego intent?
- What is the best Action now?



Reflex Response



Output Trajectory

<answer>

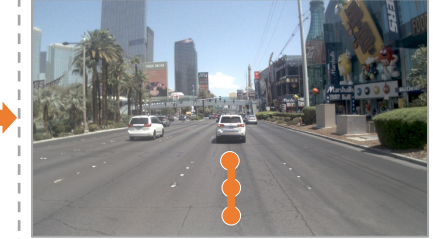
[3.340,-0.003,-0.037],

[6.082,-0.033,-0.011],

...

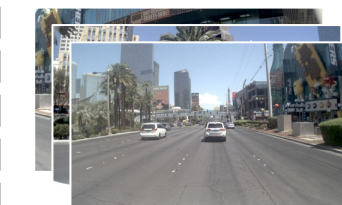
[14.130,-0.409,-0.034]

</answer>

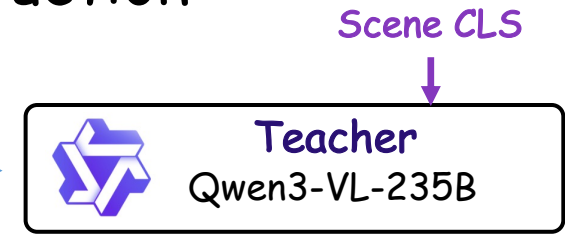


(b) Data Construction

Multi-view images

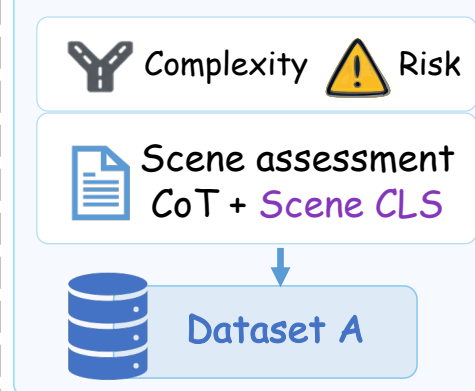


①



②

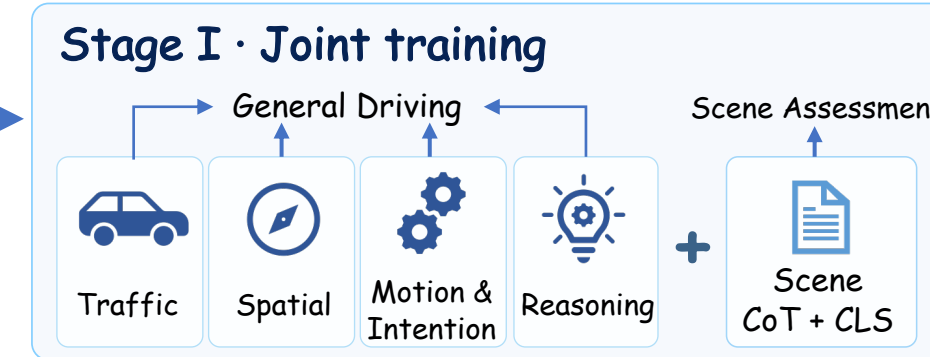
Scene assessment data



Mode-adaptive driving data



(c) Progressive SFT



Stage II · Mode-adaptive driving



(d) Reasoning-augmented RL

